

BHAVYA KANSAL

AI/ML Engineer · Deep Learning Researcher · Computer Vision & Gen AI

Patiala, Punjab, India | +91 6283332944 | kansalbhavya27@gmail.com | bhavyakansal.dev | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Deep Learning Researcher Currently in Final Year of Diploma in Computer Science Engineering at Thapar Polytechnic College while undergoing intensive 6-month industrial training at NIELIT Ropar in collaboration with IIT Ropar. Demonstrates a strong command of end-to-end AI product development — from model design, training, and optimization to full-stack deployment on cloud platforms. Has independently delivered six production-grade AI systems in field of AI in healthcare diagnostics, multimodal deepfake detection, Indian Sign Language recognition, driver safety, emotion intelligence, and agentic AI — all live and publicly accessible. Upcoming B.Tech (AI & Data Science) student at Thapar Institute of Engineering and Technology (TIET). Actively researching next-generation AI in healthcare (diabetic retinopathy, knee osteoarthritis, ethical medical AI), advanced multimodal architectures, and generative AI systems. Open to research collaborations, industry internships, and high-impact AI/ML roles.

EDUCATION

Thapar Institute of Engineering and Technology (TIET)

B.Tech in Data Science and Artificial Intelligence

July 2026 – 2029

Patiala, Punjab

Thapar Polytechnic College

Diploma in Computer Science Engineering

August 2023 – Present

Patiala, Punjab

Sunflower Model High School

Matriculation — Secondary School Certificate (10th Grade)

Completed 2022

Patiala, Punjab

EXPERIENCE

IIT Ropar × NIELIT Ropar — Joint Collaboration Program

Internship Trainee (AI/ML Industrial Training)

January 2026 – Present

Ropar, Punjab

- Undergoing intensive 6-month AI/ML industrial training program at NIELIT Ropar in active collaboration with IIT Ropar's premier innovation and research center.
- Building production-ready AI systems applying advanced deep learning, computer vision, NLP, and multimodal AI techniques acquired through mentorship-driven research exposure.
- Engaging in R&D activities focused on applied AI, technology commercialization, and real-world deployment of intelligent systems.
- Delivered multiple AI projects using multimodal agentic intelligence including multi-disease healthcare AI platform (Healthy AI) as part of program deliverables.
- Mentored by Dr. Sarwan Singh, Nikshep Paliwal, and Lovnish Verma (NIELIT Ropar).

PROJECTS

Healthy AI — Multi-Disease Medical AI Platform |

- Built and deployed a production-ready multi-module medical AI platform offering intelligent diagnostic assistance across three clinical domains: cardiovascular/Heart disease prediction, diabetes risk assessment, and brain tumor MRI classification.
- Brain tumor MRI module classifies four tumor types (Glioma, Meningioma, Pituitary, No Tumor) achieving 91.62% accuracy using a fine-tuned deep CNN architecture trained on a curated MRI dataset.
- Heart disease and diabetes modules employ ensemble ML models with clinical feature engineering, validated against standard medical benchmarks.

- Deployed on Render with a professional dark-themed Flask web interface; currently researching extension into diabetic retinopathy detection and multivariate deep learning-based knee osteoarthritis classification.
- GitHub: github.com/BhavyaKansal20/healthy-ai | Live: <https://healthyai-dlog.onrender.com/>

DeepFake Scanner — Multimodal Deepfake Detection System |

- Engineered a dual-modality deepfake detection pipeline fusing EfficientNet-B0 for visual/video frame analysis with a custom AudioMLP model extracting 26 spectral audio features to detect AI-synthesized voice clones.
- Implemented intelligent dual-model fusion scoring for final authenticity prediction, achieving strong cross-modal detection capability on benchmark deepfake datasets.
- Deployed on Hugging Face Spaces via Docker; resolved complex git LFS large-file purging, port compatibility, and Docker SDK configuration challenges during deployment.
- GitHub: github.com/BhavyaKansal20/DeepFake-Detector | Live: <https://bhavyakansal20-deepfake-scanner.hf.space/>

SignLang AI — Indian Sign Language Recognition Platform |

Designed and deployed a real-time ISL (Indian Sign Language) recognition platform using MediaPipe hand landmark extraction feeding a trained PyTorch LSTM sequence classifier.

- Integrated Google Sheets via Apps Script for session data logging, ReportLab for PDF session report generation, and a dark glassmorphism UI for accessibility-first presentation.
- Built complete Kaggle dataset pipeline for ISL gesture data curation, augmentation, and model training; deployed on Hugging Face Spaces.
- GitHub: github.com/BhavyaKansal20/SignLang-AI | Live: <https://bhavyakansal20-signlang-ai.hf.space/>

NeuroLock AI — Real-Time Emotion Analytics & Classroom Intelligence |

Built a real-time classroom emotion analytics system using a fine-tuned MobileNetV2 (alpha=1.0) architecture trained on FER datasets, with IoU-based multi-person tracking and session lifecycle management (before/during/after class phases).

- Generated automated ReportLab PDF analytics reports with Matplotlib charts per session; debugged Keras 3.x Lambda layer deserialization issues and retrained architecture from scratch on Apple M4 GPU.
- Deployed with Flask + Socket.IO backend for low-latency real-time emotion streaming to a live web dashboard.

TECHNICAL SKILLS

Languages: Python (Expert) , HTML/CSS/JS , Java (Intermediate)

ML / DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, OpenCV, MediaPipe

Deep Learning: CNN, MobileNetV2, EfficientNet-B0, LSTM, BiLSTM, GAN (Basic), Transformer architecture

Computer Vision: FaceMesh, EAR/MAR algorithms, object detection, image segmentation, feature extraction

NLP & GenAI: NLP pipelines, sequence modeling, LLM fine-tuning (basics), Retrieval-Augmented Generation (RAG), ReAct agent patterns

Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, EDA , data preprocessing pipelines

Deployment & Tools: Hugging Face Spaces, Render, Docker, Git, GitHub, VS Code, Google Colab, Kaggle

RESEARCH & ACHIEVEMENTS

Current Research Focus

- **AI in Healthcare:** Actively researching diabetic retinopathy grading using fundus image deep learning, multivariate deep learning models for knee osteoarthritis severity classification, and explainable AI (XAI) for clinical decision support in medical imaging.
- **Ethical & Responsible AI:** Investigating fairness, transparency, and accountability in medical AI systems; studying model bias detection and mitigation techniques aligned with emerging Indian AI governance frameworks.
- **Advanced Deep Learning:** Exploring next-generation architectures including Vision Transformers (ViT), diffusion models, advanced GAN variants, and self-supervised learning for low-resource scenarios.
- **Generative AI & LLMs:** Researching LLM architecture internals, parameter-efficient fine-tuning (PEFT/LoRA), RAG pipelines, and agentic AI frameworks for autonomous task execution.
- **Multimodal AI Systems:** Building expertise in cross-modal fusion (vision + audio + text), multimodal embedding spaces, and production deployment of real-time multimodal pipelines.